

SASHEEN SRI WIDANAGAMAGE

DATA ANALYTICS, DATA ENGINEERING, BI & MACHINE
LEARNING INTERN

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[LinkedIn](#) [GitHub](#) [Portfolio](#)

PROFESSIONAL SUMMARY

Data Science undergraduate at the Sri Lanka Institute of Information Technology (SLIIT), specializing in data analytics, business intelligence, data warehousing, machine learning, and AI-focused project work. Experienced in building analytical solutions using R, Python, SQL, Power BI, SSIS, SSAS, Excel OLAP, and statistical modelling. Strong project portfolio covering customer experience analytics, retail data warehousing, BI reporting, Android product development, and full-stack systems. Seeking an internship in Data Analytics, BI, Data Engineering, Data Science, Machine Learning, AI, or Big Data Analytics to apply data-driven problem solving in practical business environments.

EDUCATION

BSc (Hons) in Information Technology – Data Science Specialization

2023 – Present

Sri Lanka Institute of Information Technology (SLIIT)

- **Focus:** Data science, data analytics, business intelligence, machine learning, databases, programming, and data-driven application development.
- **Current GPA:** 3.38

Mahinda Rajapaksha College, Homagama

2014 – 2022

- Participated in Baseball, Kabaddi, Shot Put, and Discus, developing teamwork, discipline, and commitment alongside academic studies.

GCE Advanced Level Examination

- Combined Mathematics, Physics and Information & Communication Technology.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Java, Kotlin, JavaScript, PHP, HTML, CSS, XML, C/C++

Data Analytics & Statistics: Exploratory Data Analysis (EDA), data cleaning, data preprocessing, feature engineering, statistical testing, hypothesis testing, correlation analysis, ANOVA, t-test, Chi-square test

Machine Learning & AI: Logistic regression, predictive modelling, classification, model evaluation, confusion matrix, precision, recall, F1-score, ROC curve, AUC, AI/ML fundamentals

Business Intelligence: Power BI, Power BI Service, Excel PivotTables, OLAP, matrix reports, drill-down reports, drill-through reports, cascading slicers, KPI reporting

Data Engineering & Databases: SQL Server, SSIS, SSAS, MySQL, MongoDB, data warehousing, star schema, ETL pipelines, SCD Type 2, fact and dimension tables, data quality handling

Tools & Frameworks: RStudio, Jupyter Notebook, VS Code, GitHub, Android Studio, React, Vite, Spring Boot, JWT, OAuth2, Docker fundamentals, Recharts, Framer Motion

PROJECTS

Customer Service Process Analytics Using R

R, RStudio, Statistics, Logistic Regression, Data Visualization | [GitHub - sasheen-widanagamage/customer-service-process-analytics - GitHub](#)

Analyzed a dataset of **129,880 customer records across 22 features** to evaluate the relationship between service-process smoothness and customer experience, applying descriptive, inferential, and predictive analytics in R.

- Conducted ANOVA, Tukey post-hoc, Chi-square, Pearson correlation, t-test, and confidence interval analysis to statistically validate key drivers of customer satisfaction, supported by 13 professional ggplot2 visualizations.
- Built a logistic regression model achieving **ROC-AUC = 0.908**, with feature selection confirming the reduced model maintained performance comparable to the full model.
- **Technologies:** R, RStudio, tidyverse, dplyr, ggplot2, caret, pROC, data cleaning, median imputation, outlier handling, categorical encoding, logistic regression, statistical testing.

Retail Data Warehouse and ETL Solution

SQL Server, SSIS, T-SQL, Star Schema, ETL | [GitHub - sasheen-widanagamage/dwbi-retail-data-warehouse-etl](#) · [GitHub](#)

Designed and implemented an end-to-end retail data warehouse processing **301,992 source rows** from the Kaggle Retail Analysis Large Dataset, including staged source loading, dimension/fact table creation, SCD Type 2 handling, accumulating fact updates, and reject-file handling.

- Improved BI readiness of raw retail data by transforming SQL Server, CSV, and Excel sources into a star schema with structured fact and dimension tables for sales, customer, product, order profile, and date analysis.
- Technologies:** SQL Server, SQL Server Management Studio, SSIS, SQL Server Data Tools, ETL, star schema, SCD Type 2, staging database, data quality validation.

Retail Business Intelligence Analytics and Reporting

Power BI, SSAS, Excel OLAP, SQL Server | [GitHub - sasheen-widanagamage/dwbi-retail-bi-analytics-reports](#) · [GitHub](#)

- Extended the Retail Data Warehouse (above) into a full BI reporting layer by implementing an SSAS multidimensional cube, Excel OLAP analysis, and a suite of Power BI reports.
- Performed multidimensional retail sales analysis using SSAS cube measures and date/product hierarchies, delivering interactive matrix, cascading slicer, drill-down, and drill-through reports published via Power BI Service.
- Technologies:** Power BI Desktop, Power BI Service, SSAS, SQL Server, Excel PivotTables, OLAP, cube design, drill-through reports, cascading slicers.

Smart Campus Operations Hub | Group Project

React, Spring Boot, Java, MySQL, OAuth2, JWT | [Group Project – SLIIT](#)

- Developed a full-stack university operations platform with modules for facility management, resource booking, incident tickets, notifications, admin analytics, QR support, and PDF reporting.
- Implemented secure, role-based access using Google OAuth login and JWT authentication, with conflict-aware booking logic built on Spring Security, MySQL, React, Vite, Axios, Recharts, and GitHub Actions.

DailyFlow – Personal Wellness & Habit Companion

Kotlin, Android Jetpack, Material Components, SharedPreferences | <https://github.com/sasheen-widanagamage/Daily-Flow>

- Built an offline-first Android habit and wellness app featuring habit tracking, hydration reminders, sleep alarms, study/meditation timers, journal history, mood tracking, and bottom navigation, using local storage and notification scheduling.
- Technologies:** Kotlin, MVVM, XML, Material Design, SharedPreferences, AlarmManager, FragmentManager, bottom navigation.

CERTIFICATIONS

- Cisco:** Data Analytics Essentials, Introduction to Data Science, Python Essentials 1, Python Essentials 2, Introduction to Modern AI, Introduction to IoT.
- SLIIT:** AI/ML Engineer – Stage 1 and AI/ML Engineer – Stage 2.
- MongoDB:** Getting Started with MongoDB Atlas, MongoDB Atlas Security, Replication in MongoDB.
- Alison:** Python Flask Development with Docker.

Extracurricular Activities & Achievements

- Maintain a public technical portfolio of 9 GitHub repositories spanning R, T-SQL, Kotlin, Java, HTML, and PHP, documenting academic and self-directed project work for recruiter review.
- Completed 12 professional certifications in data analytics, AI/ML, Python, IoT, MongoDB, and Docker-related development through structured online and university-based credentials, reflecting continuous self-driven learning.
- Developed teamwork and discipline through Baseball, Kabaddi, Shot Put, and Discus during school, sustained alongside academic studies.

REFERENCES

Prof. Samantha Rajapaksha

Head/Department of IT

B.Sc. Eng. (Moratuwa), M.Sc.(SLIIT), Ph.D. (SLIIT).

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